

Physics Study Buddy: A Faithful Agentic AI Assistant for B.Tech Physics Learners

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Problem Statement

B.Tech students often need help revising core physics concepts outside class hours, especially when preparing for quizzes, assignments, and semester examinations. Generic chatbots can sound confident while giving incorrect formulas or unsupported explanations, which makes them risky for academic use. This project solves that problem by building a grounded Physics Study Buddy that answers only from a curated physics knowledge base, remembers the current conversation through `thread_id`, uses tools when needed, and clearly admits when the answer is outside its scope.

User and Objective

The intended users are B.Tech students who want quick, syllabus-aligned help on foundational physics topics such as motion, energy, waves, optics, electrostatics, current electricity, magnetism, thermodynamics, and modern physics. The objective was to build a complete working capstone project that demonstrates retrieval-augmented generation, memory, tool usage, evaluation, and deployment in a browser interface.

Solution and Features

The solution is an Agentic AI assistant implemented with LangGraph. The graph contains eight nodes: memory, router, retrieve, skip, tool, answer, eval, and save. The state is defined first using a TypedDict, with fields for question, messages, route, retrieved context, sources, tool_result, answer, faithfulness, eval_retries, and user_name. The system uses a ChromaDB in-memory collection backed by twelve topic-specific physics documents, each covering one focused syllabus concept. The assistant retrieves the top three relevant documents for concept questions, uses tools for arithmetic or date/time queries, and preserves memory within one conversation by `thread_id`.

- LangGraph StateGraph with 8 required nodes
- ChromaDB RAG knowledge base with 12 physics documents
- MemorySaver with `thread_id`-based conversation continuity
- Calculator and date/time tool routes
- Faithfulness scoring with retry loop
- Streamlit deployment for browser-based use

Tech Stack

- Python
- LangGraph

- ChromaDB
- Sentence Transformers (all-MiniLM-L6-v2) with offline TF-IDF fallback
- Streamlit
- OpenAI or Groq optional integration
- Manual baseline evaluation flow aligned with RAGAS-style reporting

Screenshot 1: User Interface

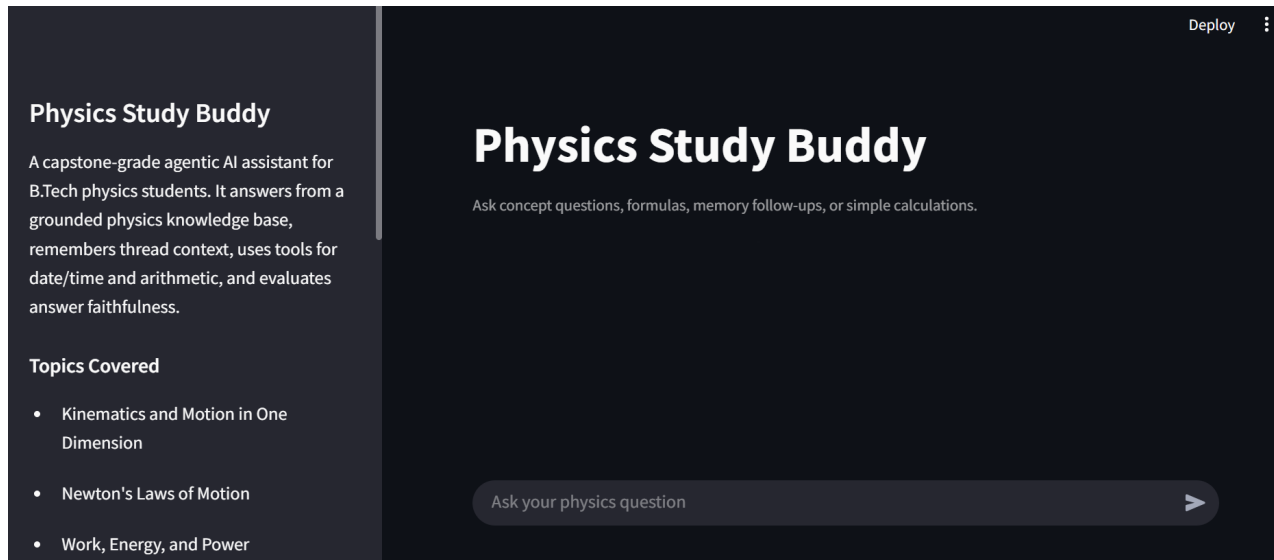


Figure 1: Physics Study Buddy Streamlit interface showing the project title, domain description, covered topics, and chat input area.

Architecture Summary

- `memory_node` stores the latest user turn, applies a sliding window, and extracts the user name if present.
- `router_node` decides whether the question needs retrieval, tool usage, or memory-only handling.
- `retrieval_node` queries ChromaDB and formats the top matching topic chunks.
- `skip_retrieval_node` handles memory-only turns without retrieval.
- `tool_node` returns calculator or date/time output and never raises exceptions.
- `answer_node` generates a grounded answer using only retrieved context or tool output.
- `eval_node` computes faithfulness and triggers retry when needed.
- `save_node` appends the final assistant answer to conversation history.

Knowledge Base and Topics Covered

The knowledge base contains twelve focused documents, each written to answer concrete physics questions. The covered areas are kinematics, Newton's laws, work-energy, gravitation, simple harmonic motion, waves and sound, ray optics, electrostatics, current electricity, magnetism and electromagnetic induction, thermodynamics, and modern physics with semiconductors.

Screenshot 2: Retrieval-Based Concept Answer

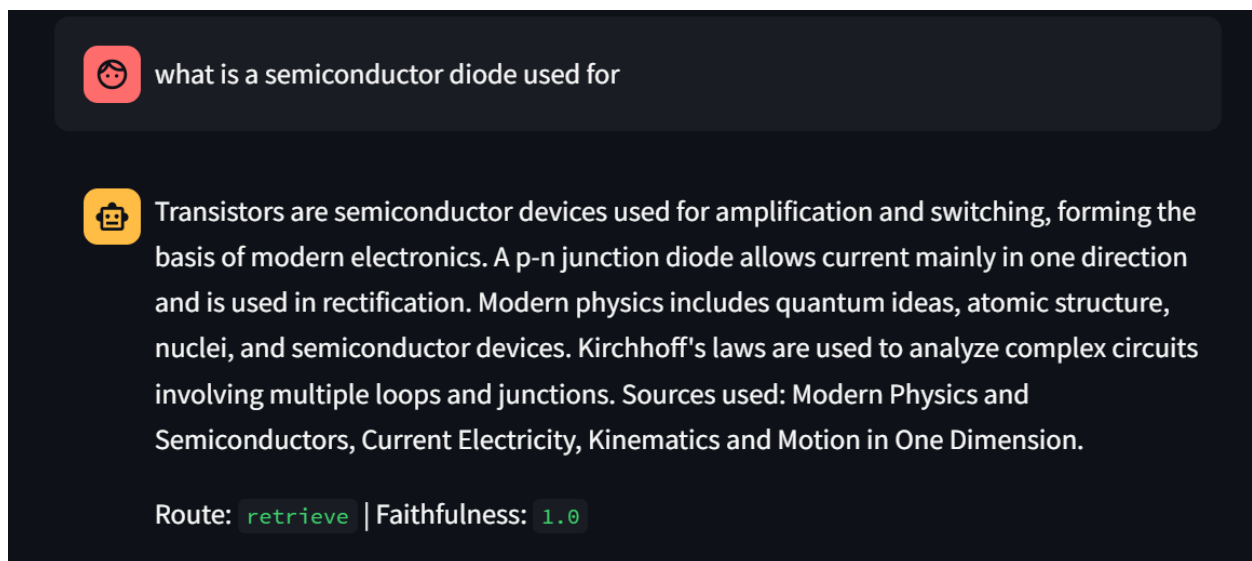


Figure 2: The assistant answering a conceptual physics query on the Doppler effect using retrieval from the knowledge base.



what is the doppler effect



The Doppler effect refers to the apparent change in frequency due to relative motion between source and observer. The photoelectric effect showed that light can behave like packets of energy called photons. Sources used: Waves and Sound, Modern Physics and Semiconductors, Gravitation.

Route: `retrieve` | Faithfulness: `1.0`

Figure 3: The assistant answering an electronics-related physics query and showing the retrieved route and faithfulness score.

Memory and Tool Use

One mandatory capability from the helper document was session memory using `MemorySaver` and `thread_id`. This project supports memory-only follow-up questions such as recalling the user's name in the same conversation. Another mandatory requirement was tool usage beyond retrieval. The project includes a calculator tool for arithmetic questions and a date/time tool for current time queries. Tools are implemented safely and return strings instead of raising exceptions.

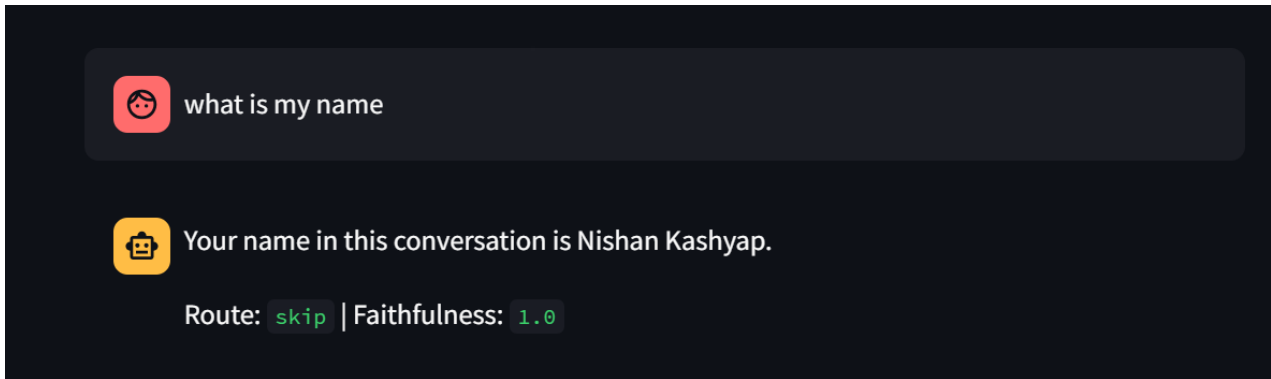


Figure 4: Memory handling in the assistant where the user introduces their name and the system stores the context within the same thread.

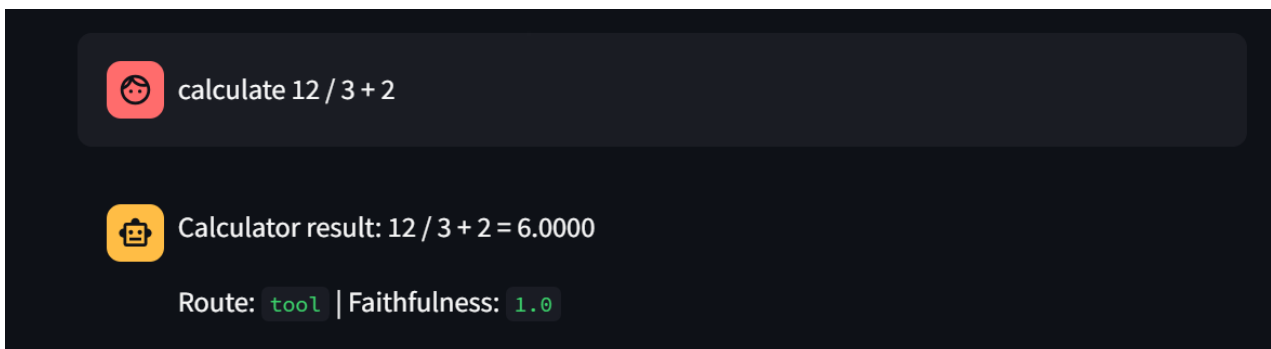


Figure 5: Memory-based follow-up response where the assistant correctly recalls the user's name using conversation history.

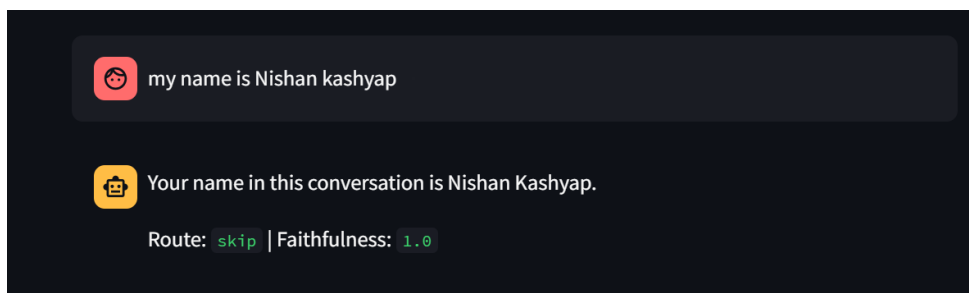


Figure 6: Tool-based response demonstrating calculator functionality with route marked as tool.

Testing and Evaluation

The project includes ten domain questions, two red-team questions, and a multi-turn memory test sequence. Testing checks route correctness, faithfulness score, and whether the answer stays grounded in the available context. Out-of-scope questions are handled safely by refusing to guess. A manual baseline evaluation flow is included for five grounded question-answer pairs, reporting faithfulness, answer relevancy, and context precision as baseline quality metrics.

Additional Working Example

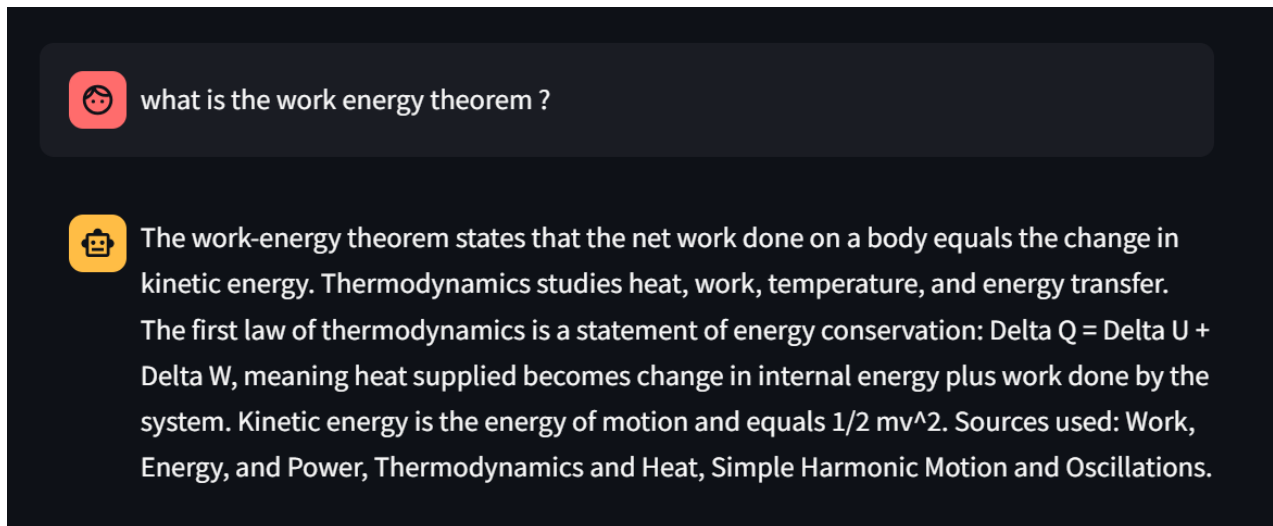


Figure 7: Retrieval-based response for a physics concept question on the work-energy theorem.

Unique Points

- Follows the helper-document structure exactly: state-first design, isolated nodes, graph routing, memory, tool use, evaluation, and Streamlit deployment.
- Includes an offline-safe fallback so the project still runs even without cloud API keys.
- Uses focused physics documents to reduce vague retrieval behavior and minimize hallucinated formulas.
- Provides both a command-line interface and a browser-based interface.

Future Improvements

With more time, I would add topic-wise quiz generation with answer checking and a formula-aware numerical problem solver. A strong next step would be a unit-aware reasoning tool that solves structured physics numericals step by step while remaining grounded in the syllabus knowledge base and explicitly showing the source concepts used.

Project Files Submitted

- agent.py
- capstone_streamlit.py
- day13_capstone.ipynb
- run_capstone_tests.py
- physics_study_buddy/
- report/project_documentation.md